

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 June 2026
Team ID	
Project Name	ShopEZ – MERN Stack E-Commerce Platform
Maximum Marks	4 Marks

Technical Architecture:

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web application for users and admins	React.js, Bootstrap, HTML, CSS
2.	Application Logic-1	Authentication and authorization	Node.js, Express.js, JWT, bcrypt.js
3.	Application Logic-2	Product management	Express.js APIs
4.	Application Logic-3	Cart and order processing	Node.js, Express.js
5.	Database	Store users, products, carts, orders	MongoDB
6.	Database Connectivity	Connect backend with database	Mongoose
7.	API Communication	Frontend-backend data exchange	Axios
8.	Infrastructure	Application deployment environment	Vite, Node.js
9.	Admin Dashboard	Manage products and orders	React.js

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and backend frameworks used	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	User authentication and protected routes	JWT Authentication, bcrypt.js
3.	Architecture	Three-tier MERN architecture	React + Express + MongoDB
4.	Availability	Accessible through web browsers anytime	Node.js Server + MongoDB
5.	Performance	Fast API responses and efficient database queries	Express.js, MongoDB